

SOHAM SENGUPTA

Java Stack-Neutral Software Engineer | Solution Architect | Backend Futureproof & AI Integration

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PROFILE

Java Stack-neutral Software Engineer and Architect with a proven track record of ahead-of-time intuitive and innovative ideas. Possesses a deep-seated ability to design high-scale and high-throughput systems, balancing architectural rigor with pragmatic execution. Driven by innovation and intuition, recently transitioned an extensive, 20-year enterprise expertise into the Generative AI era. Pioneering modern architectures utilizing Java 25, Retrieval-Augmented Generation (RAG), and advanced messaging systems. A strong advocate for "foundation-first" engineering, bridging the gap between core computer science principles and cutting-edge decentralized solutions.

CORE COMPETENCIES & TECHNICAL ARSENAL

- **Languages:** Java SE 8 through Java 25, Python.
- **Backend Futureproof & AI Integration:** Generative AI Integration, Retrieval-Augmented Generation (RAG), Model Context Protocol (MCP - Stateless/Stateful), Spring AI, LLM Elicitation Handlers.
- **Architecture & Design:** Microservices, Event-Driven Architecture (EDA), System Design (High-Scale/Throughput), Software Requirements Specification (SRS), Decoupled Architectures.
- **Frameworks & Web:** Spring Boot, Spring Cloud, Jakarta Persistence (JPA), REST APIs, OAuth.
- **Messaging & Integration:** Kafka, Mule ESB, MQTT (Mosquitto), RabbitMQ, ActiveMQ, JOLT (JSON to JSON Transformation).
- **Databases & Caching:** PostgreSQL (JSONB), MySQL, Oracle, DB2, MongoDB, Couchbase, Redis, Hazelcast.
- **Mobile & UI:** Android (Java/Kotlin), Swift (iOS), AngularJS, ZK, JS, Apache Sling, Adobe AEM.
- **DevOps, Cloud & Observability:** Docker, Linux, Maven, Prometheus/Grafana (Micrometer), ELK Stack, Splunk, Azure Vault, Zipkin. Awareness of basic cloud services (AWS/others: Object Storage, File Systems, Managed Databases, Lambda/FaaS) and Cloud Java SDKs.

PROFESSIONAL EXPERIENCE & KEY PROJECTS

Independent Practitioner, Architect & Technical Faculty | Freelance & Ideal College of Engineering

Dec 2025 – Present

- *Transitioned to independent practice and academia to prioritize family responsibilities, bringing industry-grade architectural expertise directly to the next generation of engineers while maintaining independent architectural consulting.*
- **AI Voice Bot & Institutional Chatbot:** Architected and deployed an interactive AI voice bot and an MCP-integrated chatbot backend for the institutional website to serve as an intelligent academic assistant.
- **AI-Examiner:** Architected an automated AI evaluation platform for exam scripts utilizing Spring Boot, Spring AI, PostgreSQL (JSONB), and Docker containerization.
- **Chrome Extension:** Developed and published a companion Chrome extension on the Chrome Web Store, backed by custom RAG and MCP architecture.
- **GullyDirect & NexusID:** Designed the SRS for *GullyDirect*, a hyperlocal WhatsApp-based, MQTT-driven D2C platform. Drafted the technical blueprint for *NexusID*, a universal, consent-based commerce profile and decentralized identity service.
- **Academic Leadership:** Lead web development training sessions emphasizing a "foundation-first" philosophy, covering TCP/IP, system design, algorithm analysis, Java, MySQL, MongoDB, and Docker.

Senior Specialist, Software Engineering | LTIMINDTREE

Feb 2022 – Dec 2025

- *Initiated a sabbatical to focus on family responsibilities, ultimately deciding to move out and transition fully into independent practice and academia.*
- **Citi US/CA (Project):** Served as an independent Technical Architect and Software Development Engineer (SDE).
 - Guided the team through complex legacy migrations, successfully migrating systems from Java 8 to 17, and actively paved the architectural runway for Java 25 adoption.
 - Conducted rigorous code reviews, successfully pinpointing and resolving critical architectural flaws within the ecosystem.
 - Developed several custom tools for in-house consumption utilizing JOLT (JSON to JSON Transformation) and OAuth to secure and streamline internal workflows.
- **Staples Enterprise Architecture (Project):** Operated as an enterprise architect and individual contributor.
 - Contributed heavily to supply chain business solutions, internal developer accelerators, and in-house tooling.
 - **Tech Stack:** Java, Spring Microservices, Spring Cloud, Couchbase, Redis, DB2, MySQL, PostgreSQL, Elastic Search, Splunk, Azure Vault, Kafka, ActiveMQ, Hazelcast.

Principal Architect, Web and Mobile Platforms | PAYBACK INDIA

Mar 2018 – Feb 2022

- Formulated robust architectures for new product lines, led stack modernization, and mentored development teams.
- **ICICI Mobile Platform Integration:** Acted as Chief Architect, Individual Contributor, and Lead for backend, mobile, and web applications spanning both partner and Payback environments.
 - Managed the product lifecycle from scratch to operations, successfully decoupling legacy Java systems into a modernized Spring Boot microservices architecture.
 - **Tech Stack:** Java, Spring Boot Microservices, Oracle, MongoDB, Kafka, ELK, Linux, Mule ESB, Android (Java/Kotlin), Swift (iOS), JS, Apache Sling, Adobe AEM.
- **Future Group Digital Platform Integration:** Led the digital transformation of the customer onboarding and loyalty platform integration.
 - **Tech Stack:** Java J2EE, Spring Boot Microservices, AngularJS, ZK, Android, Mule ESB.
- **Apollo Pharmacy Loyalty Platform Integration:** Architected a set of high-throughput microservices exposed for Apollo Pharmacy POS systems utilizing Spring Boot Microservices, Mule ESB, MongoDB, and Redis Cache.
- **Scan & Pay Product:** Acted as Solution Architect to integrate PAYBACK's multi-program loyalty system with UPI payments via BharatPe merchant QR codes.

Software Architect, R&D | KRYOTECH

June 2017 – Mar 2018

- Led R&D in proprietary cryptographic algorithms, successfully integrating them with commodity hardware and building viable PoCs.

Software Architect | PROPHECY SENSORLYTICS

Apr 2016 – June 2017

- Led the acquisition and transformation of sensor data transmitted by IoT gateways.
- Developed a customer-facing portal designed specifically for predictive analytics systems.

EDUCATION

- **M. Tech in Mobile Computing and Network Technology** | West Bengal University of Technology | DGPA: 8.92 (3rd rank in University)
- **B. Tech in Information Technology (Hons.)** | University of Kalyani | 80.38%
- **12th Standard** | West Bengal Council of Higher Secondary Education | 94% (8th rank in State Board)
- **10th Standard** | West Bengal Board | ~90% (38th rank in State Board)

RECOGNITIONS & PUBLICATIONS

- **Awards:** Business Excellence Award (PAYBACK, 2018-2020), CSI Young IT Professional Award (National Winner, 2008 & 2009), IBM Drona Award (2009).
- **Publications:** Multiple research papers published in Scopus and Elsevier-SCI indexed journals, focusing on novel QR code encryption, IoT, and network layer security mechanisms.

MISCELLANEOUS SKILLS & LEGACY TECHNOLOGIES

- **Legacy Languages:** C++, C.
- **Earlier Frameworks/Tools:** AWT, Swing, Struts, JSF, ZK, J2ME (JSR-82), Google Chrome Plugin development, JNI/JNA, Java Media Framework.